

- Head predicate: $h = \text{even}(X_3)$

- Background knowledge:

$$\mathbb{B} = \{\text{successor}(X_i, X_i+1) \mid X_i \in \mathbb{E}\} \cup \{\text{zero}(0)\}$$

- Positive examples:

$$\mathcal{P} = \{\text{even}(X_i) \mid X_i \bmod 2 = 0, X_i \in \mathbb{E}\}$$

- Negative examples:

$$\mathcal{N} = \{\text{even}(X_i) \mid X_i \bmod 2 = 1, X_i \in \mathbb{E}\}$$

After training, ANDRE extracted the following subrules from the propositionalized dataset:

1. $\text{even}(X_3) \text{ :- } \text{zero}(X_3).$
2. $\text{even}(X_3) \text{ :- } \text{successor}(X_2, X_3), \neg \text{even}(X_2).$
3. $\text{even}(X_3) \text{ :- } \text{successor}(X_2, X_3), \text{successor}(X_1, X_2), \text{even}(X_1).$

4. The LessThan Task:

Objective: Learn transitive and arithmetic rules that define the less-than relation between two integers using a successor-based formulation.

- Variable set: $X = \{X_1, X_2, X_3\}$
- Head variable set: $X^h = \{X_1, X_3\}$
- Auxiliary variable set: $X^a = \{X_2\}$
- Domain of constants: $\mathbb{E} = \{0, 1, \dots, 9\}$
- Body predicates:

$$b = \{\text{successor}(X_1, X_2), \text{successor}(X_2, X_3), \text{successor}(X_1, X_3), \\ \text{lessThan}(X_3, X_2), \text{lessThan}(X_2, X_1)\}$$

- Head predicate: $h = \text{lessThan}(X_3, X_1)$
- Background knowledge:

$$\mathbb{B} = \{\text{successor}(X_i, X_i+1) \mid X_i \in \mathbb{E}\} \cup \{\text{zero}(0)\}$$

- Positive examples:

$$\mathcal{P} = \{\text{lessThan}(X_i, X_j) \mid X_i < X_j, X_i, X_j \in \mathbb{E}\}$$

- Negative examples:

$$\mathcal{N} = \{\text{lessThan}(X_i, X_j) \mid X_i \geq X_j, X_i, X_j \in \mathbb{E}\}$$

After applying the propositionalization strategy, ANDRE is trained to learn rules. The test set includes integer pairs beyond 9 to evaluate generalization. The extracted subrules are:

1. $\text{lessThan}(X_3, X_1) \text{ :- } \text{successor}(X_1, X_3).$
2. $\text{lessThan}(X_3, X_1) \text{ :- } \text{lessThan}(X_3, X_2), \text{lessThan}(X_2, X_1).$

5. The Grandparent Task:

Objective: Learn the definition of a grandparent based on transitive parent relationships using both `mother` and `father` facts provided in the background knowledge.

- Variable set: $X = \{X_1, X_2, X_3\}$
- Head variable set: $X^h = \{X_1, X_3\}$

- Auxiliary variable set: $X^a = \{X_2\}$
- Body predicates:

$$b = \{\text{father}(X_1, X_2), \text{father}(X_2, X_3), \text{father}(X_1, X_3), \\ \text{mother}(X_1, X_2), \text{mother}(X_2, X_3), \text{mother}(X_1, X_3)\}$$
- Head predicate: $h = \text{grandparent}(X_1, X_3)$
- Background knowledge:

$$\mathbb{B} = \{\text{mother}(a, c), \text{mother}(c, e), \text{mother}(b, d), \text{mother}(d, f), \\ \text{father}(g, h), \text{father}(h, i), \text{father}(j, k), \text{father}(k, l), \\ \text{mother}(m, n), \text{father}(n, o), \dots\}$$
- Positive examples:

$$\mathcal{P} = \{\text{grandparent}(a, e), \text{grandparent}(b, f), \text{grandparent}(g, i), \\ \text{grandparent}(j, l), \text{grandparent}(m, o), \dots\}$$

Using the extracted dataset, ANDRE learns the following subrules:

1. $\text{grandparent}(X_1, X_3) \text{ :- mother}(X_1, X_2), \text{mother}(X_2, X_3).$
2. $\text{grandparent}(X_1, X_3) \text{ :- father}(X_1, X_2), \text{father}(X_2, X_3).$
3. $\text{grandparent}(X_1, X_3) \text{ :- mother}(X_1, X_2), \text{father}(X_2, X_3).$
4. $\text{grandparent}(X_1, X_3) \text{ :- father}(X_1, X_2), \text{mother}(X_2, X_3).$

6. The Son Task:

Objective: Learn how the son relationship can be derived from father, brother, and sister facts using transitivity and kinship inference.

- Variable set: $X = \{X_1, X_2, X_3\}$
- Head variable set: $X^h = \{X_1, X_3\}$
- Auxiliary variable set: $X^a = \{X_2\}$
- Body predicates:

$$b = \{\text{father}(X_1, X_2), \text{father}(X_2, X_3), \text{father}(X_1, X_3), \\ \text{brother}(X_1, X_2), \text{sister}(X_1, X_2), \text{son}(X_2, X_3)\}$$
- Head predicate: $h = \text{son}(X_1, X_3)$
- Background knowledge:

$$\mathbb{B} = \{\text{father}(a, b), \text{father}(a, c), \text{father}(d, e), \text{father}(d, f), \\ \text{father}(g, h), \text{father}(g, i), \text{brother}(b, c), \text{brother}(c, b), \\ \text{brother}(e, f), \text{sister}(f, e), \text{sister}(h, i), \text{sister}(i, h), \dots\}$$
- Positive examples:

$$\mathcal{P} = \{\text{son}(b, a), \text{son}(c, a), \text{son}(e, d), \dots\}$$

After applying the propositionalization strategy, ANDRE is trained and infers the following subrules:

1. $\text{son}(X_1, X_3) \text{ :- brother}(X_1, X_2), \text{son}(X_2, X_3).$
2. $\text{son}(X_1, X_3) \text{ :- father}(X_3, X_1).$

7. The Relatedness Task:

Objective: Determine whether two individuals are related, based on transitive closure over parent relationships and recursive definitions of related.

- Variable set: $X = \{X_1, X_2, X_3\}$
- Head variable set: $X^h = \{X_1, X_3\}$
- Auxiliary variable set: $X^a = \{X_2\}$
- Body predicates:

$$b = \{\text{parent}(X_1, X_2), \text{parent}(X_2, X_3), \text{parent}(X_1, X_3), \\ \text{related}(X_1, X_2), \text{related}(X_2, X_3)\}$$

- Head predicate: $h = \text{related}(X_1, X_3)$
- Background knowledge:

$$\mathbb{B} = \{\text{parent}(a, b), \text{parent}(a, c), \text{parent}(c, e), \text{parent}(c, f), \\ \text{parent}(d, c), \text{parent}(g, h), \dots\}$$

- Positive examples:

$$\mathcal{P} = \{\text{related}(a, b), \text{related}(a, c), \text{related}(a, e), \text{related}(a, f), \\ \text{related}(f, a), \text{related}(a, a), \text{related}(d, b), \text{related}(h, g), \dots\}$$

ANDRE is then trained using the propositionalized dataset. The extracted subrules are as follows:

1. $\text{related}(X_1, X_3) \text{ :- parent}(X_1, X_3).$
2. $\text{related}(X_1, X_3) \text{ :- parent}(X_3, X_1).$
3. $\text{related}(X_1, X_3) \text{ :- related}(X_3, X_1).$
4. $\text{related}(X_1, X_3) \text{ :- related}(X_1, X_2), \text{related}(X_2, X_3).$

8. The Father Task:

Objective: Infer the father relationship using background assumptions involving marriage and motherhood.

- Variable set: $X = \{X_1, X_2, X_3\}$
- Head variable set: $X^h = \{X_1, X_3\}$
- Auxiliary variable set: $X^a = \{X_2\}$
- Body predicates:

$$b = \{\text{mother}(X_1, X_2), \text{mother}(X_2, X_3), \text{mother}(X_1, X_3), \\ \text{husband}(X_1, X_2), \text{husband}(X_2, X_3), \text{husband}(X_1, X_3)\}$$

- Head predicate: $h = \text{father}(X_1, X_3)$
- Background knowledge:

$$\mathbb{B} = \{\text{aunt_of}(\text{Howard}, \text{Anne}), \text{aunt_of}(\text{Anne}, \text{Henry}), \\ \text{mother_of}(\text{Anne}, \text{Elizabeth}), \text{married_to}(\text{Henry}, \text{Anne}), \\ \text{brother_of}(\text{John}, \text{Margaret}), \text{brother_of}(\text{Henry}, \text{Louis}), \\ \text{married_to}(\text{Louis}, \text{Adele}), \text{mother_of}(\text{Adele}, \text{Philip}), \\ \text{brother_of}(\text{Philip}, \text{Margaret}), \dots\}$$

- Positive examples:

$$\mathcal{P} = \{\text{father}(\text{Louis}, \text{Philip}), \text{father}(\text{Henry}, \text{Elizabeth}), \dots\}$$

After applying the propositionalization strategy, ANDRE is trained and infers the following subrule:

1. $\text{father}(X_1, X_3) \text{ :- husband}(X_1, X_2), \text{mother}(X_2, X_3).$